

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

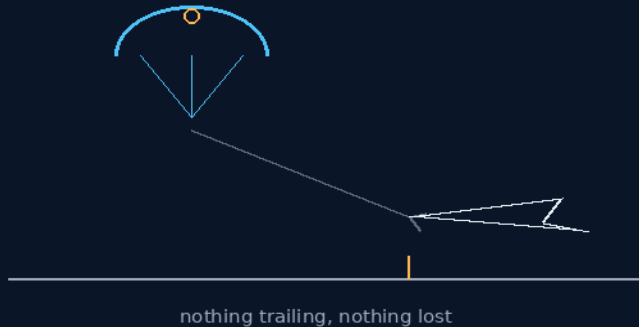
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 116

2KM TERMINAL AIRFRAME SHED & APEX BEACON RECLAMATION LOOPS

Both canopies gone two kilometres out — clean glider
beacon on the pilot chute leads the tug to the laundry



2KM GATE · SOLENOID RELEASE · APEX BEACON
THE SHIP ARRIVES AS NOTHING BUT WINGS

Terminal-recovery system — v1.0 in force

GP-IM-116

Revision v1.0 — 23 August 2026

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VETTED BATCH 17 — PASS

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 116

PHASE 116: 2KM TERMINAL AIRFRAME SHED & APEX BEACON RECLAMATION LOOPS — STATUS: CURRENT. The cloth works high and lets go two kilometres out — the ship arrives as nothing but wings. Phase 116 executes the terminal half of the cascade: a strict 2-kilometre release threshold sheds both canopy stages before the tarmac, because a trailing parachute on short final is a sail waiting for one gust of windshear off the hangars — and the 26,000 kg craft (his call, sitting 575: 22t base craft plus rocket and chutes; the 18 July 35t estimate preserved verbatim in §1) crosses the fence as a clean, predictable glider on standard commercial tires. Then the recovery loop: a ruggedized Phase 17 optical/radio beacon rides the apex crown of the pilot mini-chute, the cascade's continuous tying keeps the whole released canvas mass in one consolidated drift footprint, and the beacon talks to the Phase 49 grained network so Phase 41 Mule Tugs or drones fetch the pristine canvas back to the Deck 6 packing stations inside the 20-minute turnaround. The vet: PASS — correct aviation, orthodox recovery.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-116, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the shed-and-beacon law as seated (contents line, both bodies, the physics note, the origin — verbatim) — §2 why the windshear logic is correct (graded) — §3 the system in the gate — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 116: **2km Terminal Airframe Shed & Apex Beacon Reclamation Loops.** Executes canopy release at a strict 2-kilometer altitude threshold to eliminate tarmac windshear sail risks. Embeds a Phase 17 locator beacon in the mini-chute apex to direct automated Mule Tugs to

recover the fabric.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Runway Parachute Sail Drag & Autonomous Canopy Recovery '

Aeronautical Safety Directive: Terminal Windshear Mitigation via Disconnect

THE 2KM TERMINAL RECOVERY SHED GATE (VERBATIM)

- **Rationale:** Rejects trailing runway-level parachutes to eliminate terminal crosswind capture and localized asphalt windshear structural flips — a trailing canopy at touchdown is a sail waiting for a gust.
- **Two-Phase Solenoid Release:** Mandates the use of standard mechanical release technology to execute an disconnect of both canopy stages — the first after its high-altitude run, the second at the terminal gate.
- **The 2000m Clean Glide Profile:** Enforces canopy release at a strict 2-kilometre threshold from the tarmac, transforming the 35,000 kg craft into a clean, un-encumbered aerodynamic glider for a highly predictable touchdown on standard commercial rubber tires.

MASS LAW (sitting 575, his order verbatim): 'i was going for 15 when i was doing the lift numbers via maglev, we ended up at 22 after calculations on the base craft, we will call it 26, because of rocket and chutes'. The clean-glider figure is now 26,000 kg — the 22,000 kg base craft plus rocket and chutes. The 35,000 kg above stands preserved as the 18 July estimate — verbatim law is never rewritten. The 2-kilometre gate and the beacon reclamation loop are mass-independent; the same wing at 26t lands slower and gentler than the 35t figures it replaces — the friendly direction.

THE APEX MINI-CHUTE LOCATOR MATRIX (VERBATIM)

- **Ruggedized Core Telemetry:** Embeds an impact-resistant Phase 17 optical/radio tracking beacon into the apex crown of the initial pilot mini-chute.
- **Consolidated Drift Mapping:** Utilizes the continuous structural tying of the cascade to keep the entire released canvas mass grouped within a single, consolidated drift footprint after disconnect.
- **Autonomous Mule Reclamation:** The beacon transmits real-time coordinates to the Phase 49 grained AI network, directing Phase 41 Mule Tugs or retrieval drones to recover the pristine canvas and return it to the Deck 6 packing stations for re-racking within the 20-minute turnaround loop.

PHYSICS NOTE — PHASE 116 (KIMI AUDIT: AFFIRMED) (VERBATIM)

- **The windshear logic is correct aviation.** A deployed canopy at low altitude is a large sail on a long lever arm; gust response at the tail of a heavy glider on short final is exactly how crosswind accidents happen, so shedding drag before the boundary is the right call for a craft with strong glide performance. The recovery half is equally orthodox — every dropped rocket stage, sonobuoy, and

weather radiosonde carries a locator for the same reason: hardware that lands far away is only cheap if you can find it again.

ORIGIN OF THOUGHT — PHASE 116 (VERBATIM)

Archer, 18 July: "no, remember this is two phase so the first chute must go with standard chute release tech, the second is also released 2km out, this is a glider, windshear on tarmacs can be unpredictable. the very top of the mini chute has a tracking beacon."

Why it matters, in plain English: Gemini had drafted a fancy latch for the winches; Archer corrected the flight plan itself. A glider dragging a parachute down the approach is a kite tied to a truck — fine in calm air, a catastrophe in one gust of windshear off the hangars. So the cloth does its work high up and lets go two kilometres out, and the ship arrives as nothing but wings. And because dropped parachutes cost money, the smallest piece of the whole cascade carries a beacon — the pilot chute becomes the homing pigeon that leads a robot tug to the entire pile of laundry. Nothing lost, nothing dragging, nothing guessed.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE WINDSHEAR LOGIC, AFFIRMED AS CORRECT AVIATION: a deployed canopy at low altitude is a large sail on a long lever arm; gust response at the tail of a heavy glider on short final is exactly how crosswind accidents happen. Shedding drag before the boundary is the right call for a craft with strong glide performance — the vet said so in terms.

THE CLEAN-GLIDER PROFILE, AFFIRMED: release at a strict 2 km threshold transforms the craft into an un-encumbered aerodynamic glider — highly predictable touchdown on standard commercial rubber tires. The kinetic work is already done upstairs (Phase 114); the runway gets nothing but wings.

THE RECOVERY LOOP, AFFIRMED AS ORTHODOX: every dropped rocket stage, sonobuoy, and radiosonde carries a locator for the same reason — hardware that lands far away is only cheap if you can find it again. The smallest piece of the cascade carries the beacon; the pilot chute becomes the homing pigeon for the entire pile.

THE CONSOLIDATED FOOTPRINT, AFFIRMED: keeping the released canvas continuously tied means one drift footprint, one recovery sortie, one repack cycle — the turnaround economics hold because the laundry comes home in one bundle.

§3 — THE SYSTEM IN THE GATE

- THE GATE: strict 2-kilometre release threshold — both canopy stages gone before the boundary; the

ship crosses the fence as wings.

- THE RELEASE: standard mechanical release technology, two-phase solenoid action — first stage after its high run, second at the terminal gate.
- THE BEACON: ruggedized Phase 17 optical/radio tracker in the apex crown of the pilot mini-chute — the smallest piece carries the locator.
- THE FOOTPRINT: continuous structural tying keeps the released canvas in one consolidated drift mass.
- THE FETCH: beacon to the Phase 49 grained network; Phase 41 Mule Tugs or drones recover the canvas to Deck 6 repack — inside the 20-minute turnaround.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE RELEASE-CLEANLY DOCTRINE, COMPLETED (Phase 114's winch rule finished here): every stage lets go on schedule — nothing trails onto the tarmac.
- THE SHED-BEFORE-THE-BOUNDARY DOCTRINE (seated with the gate): windshear is not negotiated with — the sail is gone before the gust zone.
- THE LOCATOR-ON-THE-SMALLEST-PIECE DOCTRINE (the beacon): the cheap part finds the expensive pile — recovery hardware rides the pilot chute.
- THE CORRECT-THE-FLIGHT-PLAN DOCTRINE (the origin): Gemini drafted a fancy latch for the winches; Archer corrected the flight plan itself — the design review that made this phase.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Fit the release: standard two-phase solenoid mechanicals on both stages — proven dirty, wet, and cold.
- STEP 2 — Survey the gate: the 2 km threshold written into the descent profile; release rehearsed against crosswind cases.
- STEP 3 — Crown the beacon: ruggedized Phase 17 tracker potted into the pilot-chute apex; impact-rated.
- STEP 4 — Tie the footprint: continuous canvas tying verified — one drift mass, one sortie.
- STEP 5 — Drill the fetch: beacon-to-network-to-tug recovery rehearsed; Deck 6 repack timed inside

the turnaround.

- STEP 6 — Publish the ledger: release films, drift maps, turnaround clocks — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 114 — the cascade (GP-IM-114): the upstairs braking this gate finishes.
- PHASE 115 — the tension-break laws (GP-IM-115): the fuse hierarchy on every line.
- PHASE 117 — the latch (GP-IM-117): the release geometry at the rail.
- PHASE 17 — the beacon stock: the locator's origin.
- PHASE 49/41 — the grained network and the Mule Tugs: the recovery loop's movers.
- PHASE 111 — the shuttle (GP-IM-111): the turnaround this loop serves.

§8 — DO-NOT-BUILD

- DO NOT trail canvas past the 2 km gate — a parachute on short final is a sail; windshear does not negotiate.
- DO NOT improvise a 'better' release — standard mechanical release technology is the mandate (the origin records why: the flight plan, not the latch, was the fix).
- DO NOT fly without the beacon — dropped hardware is only cheap if you can find it again.
- DO NOT scatter the footprint — untied canvas is multiple sorties and a broken turnaround.

§9 — OWED

OWED: nothing — the gate, beacon, and recovery loop are complete as seated.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-116, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, ninth of the 20-chapter 'ok next 20' shift (the author's order, 23 August 2026: 'ok next 20'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders

(seated in sittings 537, 557, 561, 562, 564, 566 and 572): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 116